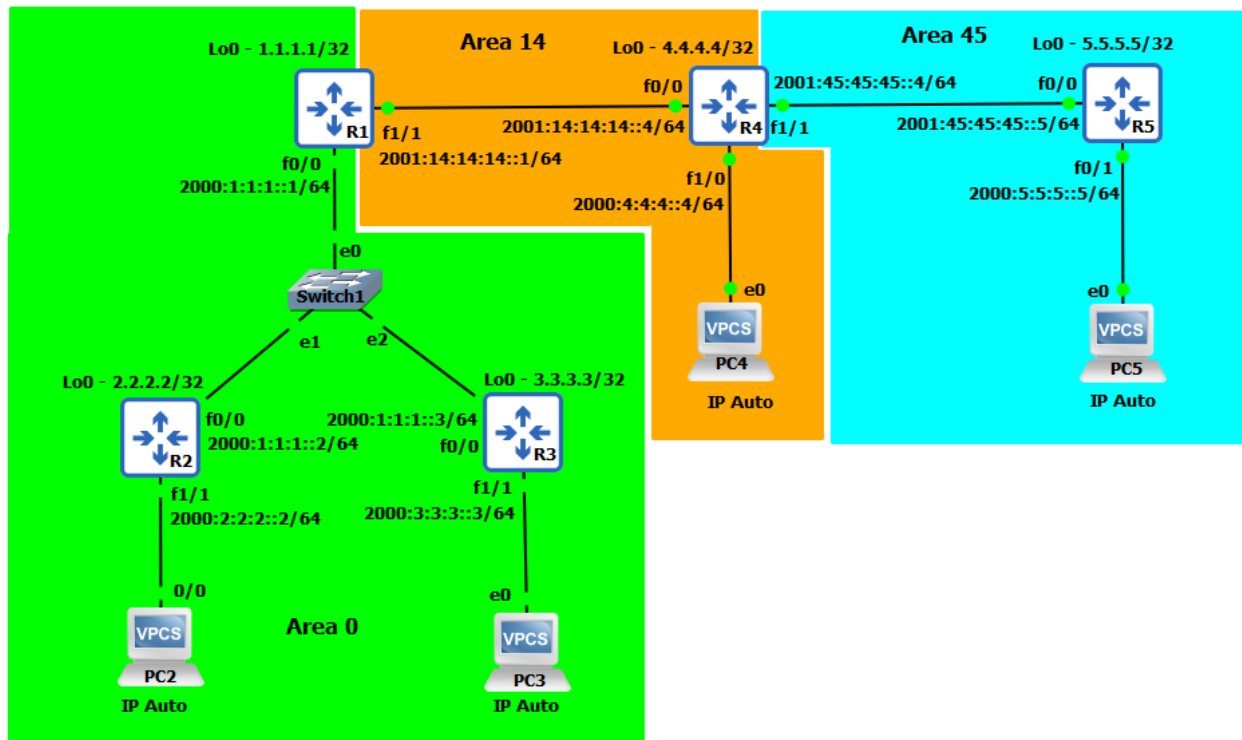


OSPFv3 IPv6 – Multi-Area

**Task**

1. Configure routers with IPv6 address as shown in topology and configure enable password as **ccna** & device access password as **cisco**.
2. Configure Multi-Area OSPFv3 (IPv6) process as per the topology.
3. Configure Virtual-link to provide communication of Area 45 with Area 0
4. Check the connectivity between routers and PCs

Task -1: Router Configuration as per the topology.**R1 Router Configuration:**

```

R1#enable
R1#config t
R1(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable password ccna
R1(config)#ipv6 unicast-routing
R1(config)#int fa0/0
R1(config-if)#ipv6 address 2000:1:1:1::1/64
R1(config-if)#description Link to Switch-1
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int fa1/1
R1(config-if)#ipv6 address 2001:14:14:14::1/64

```

```
R1(config-if)#description Link to R4 on fa0/0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int lo0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#exit
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 1 0
R1(config-line)#exit
R1(config)#line console 0
R1(config-line)#password cisco1
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 0 0
R1(config-line)#exit
R1(config)#exit
R1#
```

R2 Router Configuration:

```
R2#enable
R2#config t
R2(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#enable password ccna
R2(config)#ipv6 unicast-routing
R2(config)#int fa0/0
R2(config-if)#ipv6 address 2000:1:1:1::2/64
R2(config-if)#description Link to Switch-1
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int fa1/1
R2(config-if)#ipv6 address 2000:2:2:2::2/64
R2(config-if)#description Link to local LAN
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int lo0
R2(config-if)#ip address 2.2.2.2 255.255.255.0
R2(config-if)#exit
R2(config)#line vty 0 15
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 1 0
R2(config-line)#exit
R2(config)#line console 0
R2(config-line)#password cisco1
R2(config-line)#login
```

```
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 0 0
R2(config-line)#exit
R2(config)#exit
R2#
```

R3 Router Configuration:

```
R3#enable
R3#config t
R3(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#enable password ccna
R3(config)#ipv6 unicast-routing
R3(config)#int fa0/0
R3(config-if)#ipv6 address 2000:1:1:1::3/64
R3(config-if)#description Link to Switch-1
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int fa1/1
R3(config-if)#ipv6 address 2000:3:3:3::3/64
R3(config-if)#description Link to local LAN
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int lo0
R3(config-if)#ip address 3.3.3.3 255.255.255.0
R3(config-if)#exit
R3(config)#line vty 0 15
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 1 0
R3(config-line)#exit
R3(config)#line console 0
R3(config-line)#password cisco1
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 0 0
R3(config-line)#exit
R3(config)#exit
R3#
```

R4 Router Configuration:

```
R4#enable
R4#config t
R4(config)#hostname R4
R4(config)#no ip domain-lookup
R4(config)#enable password ccna
R4(config)#ipv6 unicast-routing
R4(config)#int fa0/0
```

```

R4(config-if)#ipv6 address 2001:14:14:14::4/64
R4(config-if)#description Link to R1 on fa1/1
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int fa1/0
R4(config-if)#ipv6 address 2000:4:4:4::4/64
R4(config-if)#description Link to local LAN
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int fa1/1
R4(config-if)#ipv6 address 2001:45:45:45::4/64
R4(config-if)#description Link to R5 on fa0/0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int lo0
R4(config-if)#ip address 4.4.4.4 255.255.255.0
R4(config-if)#exit
R4(config)#line vty 0 15
R4(config-line)#password cisco
R4(config-line)#login
R4(config-line)#logging synchronous
R4(config-line)#exec-timeout 1 0
R4(config-line)#exit
R4(config)#line console 0
R4(config-line)#password cisco1
R4(config-line)#login
R4(config-line)#logging synchronous
R4(config-line)#exec-timeout 0 0
R4(config-line)#exit
R4(config)#exit
R4#

```

R5 Router Configuration:

```

R5#
R5#enable
R5#config t
R5(config)#hostname R5
R5(config)#no ip domain-lookup
R5(config)#enable password ccna
R5(config)#ipv6 unicast-routing
R5(config)#int fa0/0
R5(config-if)#ipv6 address 2001:45:45:45::5/64
R5(config-if)#description Link to R4 on fa1/1
R5(config-if)#no shut
R5(config-if)#exit
R5(config)#int fa0/1
R5(config-if)#ipv6 address 2000:5:5:5::5/64
R5(config-if)#description Link to local LAN
R5(config-if)#no shut
R5(config-if)#exit

```

```
R5(config)#int lo0
R5(config-if)#ip address 5.5.5.5 255.255.255.0
R5(config-if)#exit
R5(config)#line vty 0 15
R5(config-line)#password cisco
R5(config-line)#login
R5(config-line)#logging synchronous
R5(config-line)#exec-timeout 1 0
R5(config-line)#exit
R5(config)#line console 0
R5(config-line)#password cisco1
R5(config-line)#login
R5(config-line)#logging synchronous
R5(config-line)#exec-timeout 0 0
R5(config-line)#exit
R5(config)#exit
R5#
```

PC2 Configuration:

```
PC2> ip auto
GLOBAL SCOPE : 2000:2:2:2:2050:79ff:fe66:6800/64
ROUTER LINK-LAYER : ca:02:07:b8:00:1d
```

PC3 Configuration:

```
PC3> ip auto
GLOBAL SCOPE : 2000:3:3:3:2050:79ff:fe66:6801/64
ROUTER LINK-LAYER : ca:03:29:c0:00:1d
```

PC4 Configuration:

```
PC4> ip auto
GLOBAL SCOPE : 2000:4:4:4:2050:79ff:fe66:6802/64
ROUTER LINK-LAYER : ca:04:26:18:00:1c
```

PC5 Configuration:

```
PC5> ip auto
GLOBAL SCOPE : 2000:5:5:5:2050:79ff:fe66:6803/64
ROUTER LINK-LAYER : ca:05:1b:d4:00:06
```

✓ Verification & Testing:**R1#sh ipv6 interface brief | exclude unassigned|down**

```
FastEthernet0/0 [up/up]
FE80::C801:25FF:FE3C:8
2000:1:1:1::1
FastEthernet1/1 [up/up]
FE80::C801:25FF:FE3C:1D
```

```

2001:14:14:14::1
Loopback0      [up/up]
R1#

```

R2#sh ipv6 interface brief | exclude unassigned|down

```

FastEthernet0/0  [up/up]
FE80::C802:7FF:FEB8:8
2000:1:1:1::2
FastEthernet1/1  [up/up]
FE80::C802:7FF:FEB8:1D
2000:2:2:2::2
Loopback0        [up/up]
R2#

```

R3#sh ipv6 interface brief | exclude unassigned|down

```

FastEthernet0/0  [up/up]
FE80::C803:29FF:FEC0:8
2000:1:1:1::3
FastEthernet1/1  [up/up]
FE80::C803:29FF:FEC0:1D
2000:3:3:3::3
Loopback0        [up/up]
R3#

```

R4#sh ipv6 interface brief | exclude unassigned|down

```

FastEthernet0/0  [up/up]
FE80::C804:26FF:FE18:8
2001:14:14:14::4
FastEthernet1/0  [up/up]
FE80::C804:26FF:FE18:1C
2000:4:4:4::4
FastEthernet1/1  [up/up]
FE80::C804:26FF:FE18:1D
2001:45:45:45::4
Loopback0        [up/up]
R4#

```

R5#sh ipv6 interface brief | exclude unassigned|down

```

FastEthernet0/0  [up/up]
FE80::C805:1BFF:FED4:8
2001:45:45:45::5
FastEthernet0/1  [up/up]
FE80::C805:1BFF:FED4:6
2000:5:5:5::5
Loopback0        [up/up]
R5#

```

Task-2: Configure Multi-Area OSPFv3 (IPv6) process as per the topology**R1 OSPFv3 Configuration:**

```
R1#config t
R1(config)#ipv6 router ospf 1
R1(config-rtr)#router-id 1.1.1.1
R1(config-rtr)#exit
R1(config)#int fa0/0
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#exit
R1(config)#int fa1/1
R1(config-if)#ipv6 ospf 1 area 14
R1(config-if)#exit
R1(config)#exit
R1#
```

R2 OSPFv3 Configuration:

```
R2#config t
R2(config)#ipv6 router ospf 2
R2(config-rtr)#router-id 2.2.2.2
R2(config-rtr)#exit
R2(config)#int fa0/0
R2(config-if)#ipv6 ospf 2 area 0
R2(config-if)#exit
R2(config)#int fa1/1
R2(config-if)#ipv6 ospf 2 area 0
R2(config-if)#exit
R2(config)#exit
R2#
```

R3 OSPFv3 Configuration:

```
R3#config t
R3(config)#ipv6 router ospf 3
R3(config-rtr)#router-id 3.3.3.3
R3(config-rtr)#exit
R3(config)#int fa0/0
R3(config-if)#ipv6 ospf 3 area 0
R3(config-if)#exit
R3(config)#int fa1/1
R3(config-if)#ipv6 ospf 3 area 0
R3(config-if)#exit
R3(config)#exit
R3#
```

R4 OSPFv3 Configuration:

```
R4#config t
R4(config)#ipv6 router ospf 4
```

```

R4(config-rtr)#router-id 4.4.4.4
R4(config-rtr)#exit
R4(config)#int fa0/0
R4(config-if)#ipv6 ospf 4 area 14
R4(config-if)#exit
R4(config)#int fa1/1
R4(config-if)#ipv6 ospf 4 area 45
R4(config-if)#exit
R4(config)#int fa1/0
R4(config-if)#ipv6 ospf 4 area 14
R4(config-if)#exit
R4(config)#exit
R4#

```

R5 OSPFv3 Configuration:

```

R5#config t
R5(config)#ipv6 router ospf 5
R5(config-rtr)#router-id 5.5.5.5
R5(config-rtr)#exit
R5(config)#int fa0/0
R5(config-if)#ipv6 ospf 5 area 45
R5(config-if)#exit
R5(config)#int fa0/1
R5(config-if)#ipv6 ospf 5 area 45
R5(config-if)#exit
R5(config)#exit
R5#

```

✓ Verification & Testing:

R1#sh ipv6 protocols

```

IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "ND"
IPv6 Routing Protocol is "ospf 1"

```

Router ID 1.1.1.1

```

Area border router
Number of areas: 2 normal, 0 stub, 0 nssa
Interfaces (Area 0):
  FastEthernet0/0
Interfaces (Area 14):
  FastEthernet1/1
Redistribution:
  None
R1#

```

R2#sh ipv6 protocols

```

IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "ND"
IPv6 Routing Protocol is "ospf 2"

```

Router ID 2.2.2.2

Number of areas: 1 normal, 0 stub, 0 nssa
Interfaces (Area 0):
 FastEthernet1/1
 FastEthernet0/0
Redistribution:
 None
R2#

R3#sh ipv6 protocols

IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "ND"
IPv6 Routing Protocol is "ospf 3"

Router ID 3.3.3.3

Number of areas: 1 normal, 0 stub, 0 nssa
Interfaces (Area 0):
 FastEthernet1/1
 FastEthernet0/0
Redistribution:
 None
R3#

R4#sh ipv6 protocols

IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "ND"
IPv6 Routing Protocol is "ospf 4"

Router ID 4.4.4.4

Number of areas: 2 normal, 0 stub, 0 nssa
Interfaces (Area 14):
 FastEthernet1/0
 FastEthernet0/0
Interfaces (Area 45):
 FastEthernet1/1
Redistribution:
 None
R4#

R5#sh ipv6 protocols

IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "ND"
IPv6 Routing Protocol is "ospf 5"

Router ID 5.5.5.5

Number of areas: 1 normal, 0 stub, 0 nssa
Interfaces (Area 45):
 FastEthernet0/1
 FastEthernet0/0
Redistribution:
 None
R5#

R1#sh ipv6 ospf interface

FastEthernet0/0 is up, line protocol is up

Link Local Address FE80::C801:25FF:FE3C:8, Interface ID 3
 Area 0, Process ID 1, Instance ID 0, Router ID 1.1.1.1
Network Type BROADCAST, Cost: 1
 Transmit Delay is 1 sec, State DROTHER, Priority 1
 Designated Router (ID) 3.3.3.3, local address FE80::C803:29FF:FEC0:8
 Backup Designated router (ID) 2.2.2.2, local address FE80::C802:7FF:FEB8:8
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
 Hello due in 00:00:06
 Graceful restart helper support enabled
 Index 1/1/1, flood queue length 0
 Next 0x0(0)/0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 2
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 2, Adjacent neighbor count is 2
 Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
 Adjacent with neighbor 3.3.3.3 (Designated Router)
 Suppress hello for 0 neighbor(s)

FastEthernet1/1 is up, line protocol is up

Link Local Address FE80::C801:25FF:FE3C:1D, Interface ID 6
 Area 14, Process ID 1, Instance ID 0, Router ID 1.1.1.1
Network Type BROADCAST, Cost: 1
 Transmit Delay is 1 sec, State BDR, Priority 1
 Designated Router (ID) 4.4.4.4, local address FE80::C804:26FF:FE18:8
 Backup Designated router (ID) 1.1.1.1, local address FE80::C801:25FF:FE3C:1D
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
 Hello due in 00:00:04
 Graceful restart helper support enabled
 Index 1/1/2, flood queue length 0
 Next 0x0(0)/0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 2
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 1, Adjacent neighbor count is 1
 Adjacent with neighbor 4.4.4.4 (Designated Router)
 Suppress hello for 0 neighbor(s)

R1#

R2#sh ipv6 ospf interface

FastEthernet1/1 is up, line protocol is up

Link Local Address FE80::C802:7FF:FEB8:1D, Interface ID 6
 Area 0, Process ID 2, Instance ID 0, Router ID 2.2.2.2
Network Type BROADCAST, Cost: 1
 Transmit Delay is 1 sec, State DR, Priority 1
 Designated Router (ID) 2.2.2.2, local address FE80::C802:7FF:FEB8:1D
 No backup designated router on this network
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
 Hello due in 00:00:07
 Graceful restart helper support enabled
 Index 1/2/2, flood queue length 0
 Next 0x0(0)/0x0(0)/0x0(0)
 Last flood scan length is 0, maximum is 0

Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)

FastEthernet0/0 is up, line protocol is up

Link Local Address FE80::C802:7FF:FEB8:8, Interface ID 3
Area 0, Process ID 2, Instance ID 0, Router ID 2.2.2.2

Network Type BROADCAST, Cost: 1

Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 3.3.3.3, local address FE80::C803:29FF:FEC0:8
Backup Designated router (ID) 2.2.2.2, local address FE80::C802:7FF:FEB8:8
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:07
Graceful restart helper support enabled
Index 1/1/1, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 0, maximum is 2
Last flood scan time is 4 msec, maximum is 4 msec
Neighbor Count is 2, Adjacent neighbor count is 2
Adjacent with neighbor 1.1.1.1
Adjacent with neighbor 3.3.3.3 (Designated Router)
Suppress hello for 0 neighbor(s)

R2#

R3#sh ipv6 ospf interface**FastEthernet1/1 is up, line protocol is up**

Link Local Address FE80::C803:29FF:FEC0:1D, Interface ID 6
Area 0, Process ID 3, Instance ID 0, Router ID 3.3.3.3
Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 3.3.3.3, local address FE80::C803:29FF:FEC0:1D
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:08
Graceful restart helper support enabled
Index 1/2/2, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 0, maximum is 0
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)

FastEthernet0/0 is up, line protocol is up

Link Local Address FE80::C803:29FF:FEC0:8, Interface ID 3
Area 0, Process ID 3, Instance ID 0, Router ID 3.3.3.3
Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 3.3.3.3, local address FE80::C803:29FF:FEC0:8
Backup Designated router (ID) 2.2.2.2, local address FE80::C802:7FF:FEB8:8
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:00

Graceful restart helper support enabled
 Index 1/1/1, flood queue length 0
 Next 0x0(0)/0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 7
 Last flood scan time is 0 msec, maximum is 4 msec
 Neighbor Count is 2, Adjacent neighbor count is 2
 Adjacent with neighbor 1.1.1.1
 Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
 Suppress hello for 0 neighbor(s)
 R3#

R4#sh ipv6 ospf interface

FastEthernet1/0 is up, line protocol is up

Link Local Address FE80::C804:26FF:FE18:1C, Interface ID 5
 Area 14, Process ID 4, Instance ID 0, Router ID 4.4.4.4

Network Type BROADCAST, Cost: 1

Transmit Delay is 1 sec, State DR, Priority 1
 Designated Router (ID) 4.4.4.4, local address FE80::C804:26FF:FE18:1C
 No backup designated router on this network
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
 Hello due in 00:00:03
 Graceful restart helper support enabled
 Index 1/2/3, flood queue length 0
 Next 0x0(0)/0x0(0)/0x0(0)
 Last flood scan length is 0, maximum is 0
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 0, Adjacent neighbor count is 0
 Suppress hello for 0 neighbor(s)

FastEthernet0/0 is up, line protocol is up

Link Local Address FE80::C804:26FF:FE18:8, Interface ID 3
 Area 14, Process ID 4, Instance ID 0, Router ID 4.4.4.4

Network Type BROADCAST, Cost: 1

Transmit Delay is 1 sec, State DR, Priority 1
 Designated Router (ID) 4.4.4.4, local address FE80::C804:26FF:FE18:8
 Backup Designated router (ID) 1.1.1.1, local address FE80::C801:25FF:FE3C:1D

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

 Hello due in 00:00:05
 Graceful restart helper support enabled
 Index 1/1/1, flood queue length 0
 Next 0x0(0)/0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 5
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 1, Adjacent neighbor count is 1
 Adjacent with neighbor 1.1.1.1 (Backup Designated Router)
 Suppress hello for 0 neighbor(s)

FastEthernet1/1 is up, line protocol is up

Link Local Address FE80::C804:26FF:FE18:1D, Interface ID 6
 Area 45, Process ID 4, Instance ID 0, Router ID 4.4.4.4

Network Type BROADCAST, Cost: 1

Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 5.5.5.5, local address FE80::C805:1BFF:FED4:8
Backup Designated router (ID) 4.4.4.4, local address FE80::C804:26FF:FE18:1D
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:03
Graceful restart helper support enabled
Index 1/1/2, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 2, maximum is 2
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
Adjacent with neighbor 5.5.5.5 (Designated Router)
Suppress hello for 0 neighbor(s)

R4#

R5#sh ipv6 ospf interface

FastEthernet0/1 is up, line protocol is up

Link Local Address FE80::C805:1BFF:FED4:6, Interface ID 4
Area 45, Process ID 5, Instance ID 0, Router ID 5.5.5.5
Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 5.5.5.5, local address FE80::C805:1BFF:FED4:6
No backup designated router on this network

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Hello due in 00:00:09
Graceful restart helper support enabled
Index 1/2/2, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 0, maximum is 0
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)

FastEthernet0/0 is up, line protocol is up

Link Local Address FE80::C805:1BFF:FED4:8, Interface ID 3
Area 45, Process ID 5, Instance ID 0, Router ID 5.5.5.5
Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 5.5.5.5, local address FE80::C805:1BFF:FED4:8
Backup Designated router (ID) 4.4.4.4, local address FE80::C804:26FF:FE18:1D

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Hello due in 00:00:04
Graceful restart helper support enabled
Index 1/1/1, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 2, maximum is 4
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
Adjacent with neighbor 4.4.4.4 (Backup Designated Router)
Suppress hello for 0 neighbor(s)

R5#

R1#sh ipv6 ospf neighbor

OSPFv3 Router with ID (1.1.1.1) (Process ID 1)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
2.2.2.2	1	FULL/BDR	00:00:31	3	FastEthernet0/0
3.3.3.3	1	FULL/DR	00:00:37	3	FastEthernet0/0
4.4.4.4	1	FULL/DR	00:00:38	3	FastEthernet1/1

R1#

R2#sh ipv6 ospf neighbor

OSPFv3 Router with ID (2.2.2.2) (Process ID 2)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
1.1.1.1	1	FULL/DROTHER	00:00:31	3	FastEthernet0/0
3.3.3.3	1	FULL/DR	00:00:34	3	FastEthernet0/0

R2#

R3#sh ipv6 ospf neighbor

OSPFv3 Router with ID (3.3.3.3) (Process ID 3)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
1.1.1.1	1	FULL/DROTHER	00:00:36	3	FastEthernet0/0
2.2.2.2	1	FULL/BDR	00:00:33	3	FastEthernet0/0

R3#

R4#sh ipv6 ospf neighbor

OSPFv3 Router with ID (4.4.4.4) (Process ID 4)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
1.1.1.1	1	FULL/BDR	00:00:33	6	FastEthernet0/0
5.5.5.5	1	FULL/DR	00:00:31	3	FastEthernet1/1

R4#

R5#sh ipv6 ospf neighbor

OSPFv3 Router with ID (5.5.5.5) (Process ID 5)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
4.4.4.4	1	FULL/BDR	00:00:32	6	FastEthernet0/0

R5#

R1#sh ipv6 route ospf

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

O 2000:2:2:2::/64 [110/2]

via FE80::C802:7FF:FE8B:8, FastEthernet0/0

O 2000:3:3:3::/64 [110/2]

via FE80::C803:29FF:FE0C:8, FastEthernet0/0

O 2000:4:4:4::/64 [110/2]

via FE80::C804:26FF:FE18:8, FastEthernet1/1

R1#

R2#sh ipv6 route ospf

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

O 2000:3:3:3::/64 [110/2]

via FE80::C803:29FF:FE0C:8, FastEthernet0/0

OI 2000:4:4:4::/64 [110/3]

via FE80::C801:25FF:FE3C:8, FastEthernet0/0

OI 2001:14:14:14::/64 [110/2]

via FE80::C801:25FF:FE3C:8, FastEthernet0/0

R2#

R3#sh ipv6 route ospf

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

O 2000:2:2:2::/64 [110/2]

via FE80::C802:7FF:FE8B:8, FastEthernet0/0

OI 2000:4:4:4::/64 [110/3]

via FE80::C801:25FF:FE3C:8, FastEthernet0/0

OI 2001:14:14:14::/64 [110/2]

via FE80::C801:25FF:FE3C:8, FastEthernet0/0

R3#

R4#sh ipv6 route ospf

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

```
OI 2000:1:1:1::/64 [110/2]
  via FE80::C801:25FF:FE3C:1D, FastEthernet0/0
OI 2000:2:2:2::/64 [110/3]
  via FE80::C801:25FF:FE3C:1D, FastEthernet0/0
OI 2000:3:3:3::/64 [110/3]
  via FE80::C801:25FF:FE3C:1D, FastEthernet0/0
O 2000:5:5:5::/64 [110/2]
  via FE80::C805:1BFF:FED4:8, FastEthernet1/1
R4#
```

R5#sh ipv6 route ospf

IPv6 Routing Table - default - 5 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

----- No OSPF Routes -----

R5#

Ping PCs of all other routers from R1

R1#ping ipv6 2000:2:2:2:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6800, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/35/92 ms

R1#ping ipv6 2000:3:3:3:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6801, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 24/38/84 ms

R1#ping ipv6 2000:4:4:4:2050:79ff:fe66:6802

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:4:4:4:2050:79FF:FE66:6802, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 24/44/112 ms

R1#ping ipv6 2000:5:5:5:2050:79ff:fe66:6803

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:5:5:5:2050:79FF:FE66:6803, timeout is 2 seconds:

% No valid route for destination
Success rate is 0 percent (0/1)

R1#

Ping PCs of all other routers from R2

R2#ping ipv6 2000:3:3:3:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6801, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/41/104 ms

R2#ping ipv6 2000:4:4:4:2050:79ff:fe66:6802

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:4:4:4:2050:79FF:FE66:6802, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/56/96 ms

R2#ping ipv6 2000:5:5:5:2050:79ff:fe66:6803

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:5:5:5:2050:79FF:FE66:6803, timeout is 2 seconds:

% No valid route for destination
Success rate is 0 percent (0/1)

R2#

Ping PCs of all other routers from R3

R3#ping ipv6 2000:2:2:2:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6800, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 24/35/76 ms

R3#ping ipv6 2000:4:4:4:2050:79ff:fe66:6802

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:4:4:4:2050:79FF:FE66:6802, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 48/67/136 ms

R3#ping ipv6 2000:5:5:5:2050:79ff:fe66:6803

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:5:5:5:2050:79FF:FE66:6803, timeout is 2 seconds:

% No valid route for destination
Success rate is 0 percent (0/1)

R3#

Ping PCs of all other routers from R4

R4#ping ipv6 2000:2:2:2:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6800, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 32/56/100 ms

R4#ping ipv6 2000:3:3:3:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6801, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/54/72 ms

R4#ping ipv6 2000:5:5:5:2050:79ff:fe66:6803

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:5:5:5:2050:79FF:FE66:6803, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/46/132 ms

R4#

Ping PCs of all other routers from R5

R5#ping ipv6 2000:2:2:2:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6800, timeout is 2 seconds:

% No valid route for destination

Success rate is 0 percent (0/1)

R5#ping ipv6 2000:3:3:3:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6801, timeout is 2 seconds:

% No valid route for destination

Success rate is 0 percent (0/1)

R5#ping ipv6 2000:4:4:4:2050:79ff:fe66:6802

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:4:4:4:2050:79FF:FE66:6802, timeout is 2 seconds:

% No valid route for destination

Success rate is 0 percent (0/1)

R5#

Task 3: Configure Virtual-link to provide communication between Area 45 and Area 0.

R1 Router Configuration:

R1#config t

R1(config)#ipv6 router ospf 1

R1(config-rtr)#area 14 virtual-link 4.4.4.4

R1(config-rtr)#exit

```
R1(config)#exit
R1#
```

R4 Router Configuration:

```
R4#config t
R4(config)#ipv6 router ospf 4
R4(config-rtr)#area 14 virtual-link 1.1.1.1
R4(config-rtr)##exit
R4(config-rtr)#exit
R4(config)#exit
R4#
```

***Oct 20 22:15:00.503: %OSPFv3-5-ADJCHG: Process 4, Nbr 1.1.1.1 on OSPFv3_VL0 from LOADING to FULL, Loading Done**

✓ Verification & Testing:

R1#sh ipv6 ospf neighbor

OSPFv3 Router with ID (1.1.1.1) (Process ID 1)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
4.4.4.4	0	FULL/ -	-	13	OSPFv3_VL0
2.2.2.2	1	FULL/BDR	00:00:31	3	FastEthernet0/0
3.3.3.3	1	FULL/DR	00:00:31	3	FastEthernet0/0
4.4.4.4	1	FULL/DR	00:00:32	3	FastEthernet1/1

R1#

R4#sh ipv6 ospf neighbor

OSPFv3 Router with ID (4.4.4.4) (Process ID 4)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
1.1.1.1	0	FULL/ -	-	13	OSPFv3_VL0
1.1.1.1	1	FULL/BDR	00:00:35	6	FastEthernet0/0
5.5.5.5	1	FULL/DR	00:00:39	3	FastEthernet1/1

R4#

R1#sh ipv6 route ospf

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

- O 2000:2:2::/64 [110/2]
via FE80::C802:7FF:FEB8:8, FastEthernet0/0
- O 2000:3:3::/64 [110/2]
via FE80::C803:29FF:FEC0:8, FastEthernet0/0

O 2000:4:4:4::/64 [110/2]
via FE80::C804:26FF:FE18:8, FastEthernet1/1
O 2000:4:4:4::4/128 [110/1]
via FE80::C804:26FF:FE18:8, FastEthernet1/1
OI 2000:5:5:5::/64 [110/3]
via FE80::C804:26FF:FE18:8, FastEthernet1/1
OI 2001:45:45:45::/64 [110/2]
via FE80::C804:26FF:FE18:8, FastEthernet1/1
R1#

R2#sh ipv6 route ospf

IPv6 Routing Table - default - 12 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

O 2000:3:3:3::/64 [110/2]
via FE80::C803:29FF:FEC0:8, FastEthernet0/0
OI 2000:4:4:4::/64 [110/3]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2000:4:4:4::4/128 [110/2]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2000:5:5:5::/64 [110/4]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2001:14:14:14::/64 [110/2]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2001:14:14:14::1/128 [110/1]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2001:45:45:45::/64 [110/3]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
R2#

R3#sh ipv6 route ospf

IPv6 Routing Table - default - 12 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

O 2000:2:2:2::/64 [110/2]
via FE80::C802:7FF:FEB8:8, FastEthernet0/0
OI 2000:4:4:4::/64 [110/3]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2000:4:4:4::4/128 [110/2]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2000:5:5:5::/64 [110/4]

```

via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2001:14:14:14::/64 [110/2]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2001:14:14:14::1/128 [110/1]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
OI 2001:45:45:45::/64 [110/3]
via FE80::C801:25FF:FE3C:8, FastEthernet0/0
R3#

```

R4#sh ipv6 route ospf

IPv6 Routing Table - default - 12 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

```

O 2000:1:1:1::/64 [110/2]
via FE80::C801:25FF:FE3C:1D, FastEthernet0/0
O 2000:2:2:2::/64 [110/3]
via FE80::C801:25FF:FE3C:1D, FastEthernet0/0
O 2000:3:3:3::/64 [110/3]
via FE80::C801:25FF:FE3C:1D, FastEthernet0/0
O 2000:5:5:5::/64 [110/2]
via FE80::C805:1BFF:FED4:8, FastEthernet1/1
O 2001:14:14:14::1/128 [110/1]
via FE80::C801:25FF:FE3C:1D, FastEthernet0/0
R4#

```

R5#sh ipv6 route ospf

IPv6 Routing Table - default - 12 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

```

OI 2000:1:1:1::/64 [110/3]
via FE80::C804:26FF:FE18:1D, FastEthernet0/0
OI 2000:2:2:2::/64 [110/4]
via FE80::C804:26FF:FE18:1D, FastEthernet0/0
OI 2000:3:3:3::/64 [110/4]
via FE80::C804:26FF:FE18:1D, FastEthernet0/0
OI 2000:4:4:4::/64 [110/2]
via FE80::C804:26FF:FE18:1D, FastEthernet0/0
OI 2000:4:4:4::4/128 [110/1]
via FE80::C804:26FF:FE18:1D, FastEthernet0/0
OI 2001:14:14:14::/64 [110/2]
via FE80::C804:26FF:FE18:1D, FastEthernet0/0

```

OI 2001:14:14:14::1/128 [110/2]
 via FE80::C804:26FF:FE18:1D, FastEthernet0/0
 R5#

R1#sh ipv6 ospf virtual-links

OSPFv3 Router with ID (1.1.1.1) (Process ID 1)

Virtual Link OSPFv3_VL0 to router 4.4.4.4 is up

Interface ID 13, IPv6 address 2000:4:4:4::4

Run as demand circuit

DoNotAge LSA allowed.

Transit area 14, via interface FastEthernet1/1, Cost of using 1

Transmit Delay is 1 sec, State POINT_TO_POINT,

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Adjacency State FULL (Hello suppressed)

Index 1/3/4, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

R1#

R4#sh ipv6 ospf virtual-links

OSPFv3 Router with ID (4.4.4.4) (Process ID 4)

Virtual Link OSPFv3_VL0 to router 1.1.1.1 is up

Interface ID 13, IPv6 address 2001:14:14:14::1

Run as demand circuit

DoNotAge LSA allowed.

Transit area 14, via interface FastEthernet0/0, Cost of using 1

Transmit Delay is 1 sec, State POINT_TO_POINT,

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Adjacency State FULL (Hello suppressed)

Index 1/1/3, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

R4#

Ping R5's PC IP from R1

R1#ping ipv6 2000:5:5:5:2050:79ff:fe66:6803

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:5:5:5:2050:79FF:FE66:6803, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 48/51/56 ms

R1#

Ping R5's PC IP from R2**R2#ping ipv6 2000:5:5:5:2050:79ff:fe66:6803**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:5:5:5:2050:79FF:FE66:6803, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 68/77/92 ms

R2#

Ping R5's PC IP from R3**R3#ping ipv6 2000:5:5:5:2050:79ff:fe66:6803**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:5:5:5:2050:79FF:FE66:6803, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 68/73/80 ms

R3#

Ping R5's PC IP from R4**R4#ping ipv6 2000:5:5:5:2050:79ff:fe66:6803**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:5:5:5:2050:79FF:FE66:6803, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 24/27/28 ms

R4#

Ping PCs of all other routers from R5**R5#ping ipv6 2000:2:2:2:2050:79ff:fe66:6800**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6800, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 64/82/124 ms

R5#

R5#ping ipv6 2000:3:3:3:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6801, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 64/70/72 ms

R5#

R5#ping ipv6 2000:4:4:4:2050:79ff:fe66:6802

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:4:4:4:2050:79FF:FE66:6802, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/27/36 ms

R5#

Ping all other PCs from PC2**PC2> ping 2000:3:3:3:2050:79ff:fe66:6801**

2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=1 ttl=60 time=60.156 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=2 ttl=60 time=43.244 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=3 ttl=60 time=41.113 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=4 ttl=60 time=43.013 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=5 ttl=60 time=43.154 ms

PC2> ping 2000:4:4:4:2050:79ff:fe66:6802

2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=1 ttl=58 time=72.072 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=2 ttl=58 time=63.098 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=3 ttl=58 time=62.048 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=4 ttl=58 time=62.190 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=5 ttl=58 time=64.098 ms

PC2> ping 2000:5:5:5:2050:79ff:fe66:6803

2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=1 ttl=56 time=92.059 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=2 ttl=56 time=84.988 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=3 ttl=56 time=84.049 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=4 ttl=56 time=85.052 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=5 ttl=56 time=84.020 ms

Ping all other PCs from PC3**PC3> ping 2000:2:2:2:2050:79ff:fe66:6800**

2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=1 ttl=60 time=38.176 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=2 ttl=60 time=42.091 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=3 ttl=60 time=41.092 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=4 ttl=60 time=41.287 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=5 ttl=60 time=42.067 ms

PC3> ping 2000:4:4:4:2050:79ff:fe66:6802

2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=1 ttl=58 time=72.051 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=2 ttl=58 time=63.133 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=3 ttl=58 time=64.922 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=4 ttl=58 time=63.160 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=5 ttl=58 time=63.046 ms

PC3> ping 2000:5:5:5:2050:79ff:fe66:6803

2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=1 ttl=56 time=85.962 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=2 ttl=56 time=85.024 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=3 ttl=56 time=85.976 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=4 ttl=56 time=82.910 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=5 ttl=56 time=86.059 ms

Ping all other PCs from PC4**PC4> ping 2000:2:2:2:2050:79ff:fe66:6800**

2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=1 ttl=58 time=100.976 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=2 ttl=58 time=64.038 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=3 ttl=58 time=64.061 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=4 ttl=58 time=65.070 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=5 ttl=58 time=64.973 ms

PC4> ping 2000:3:3:3:2050:79ff:fe66:6801

2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=1 ttl=58 time=58.238 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=2 ttl=58 time=64.093 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=3 ttl=58 time=62.057 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=4 ttl=58 time=65.195 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=5 ttl=58 time=62.968 ms

PC4> ping 2000:5:5:5:2050:79ff:fe66:6803

2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=1 ttl=60 time=49.095 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=2 ttl=60 time=42.109 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=3 ttl=60 time=41.131 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=4 ttl=60 time=41.993 ms
2000:5:5:5:2050:79ff:fe66:6803 icmp6_seq=5 ttl=60 time=42.233 ms

PC5> ping 2000:2:2:2:2050:79ff:fe66:6800

2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=1 ttl=56 time=86.039 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=2 ttl=56 time=84.044 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=3 ttl=56 time=84.993 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=4 ttl=56 time=83.090 ms
2000:2:2:2:2050:79ff:fe66:6800 icmp6_seq=5 ttl=56 time=85.989 ms

PC5> ping 2000:3:3:3:2050:79ff:fe66:6801

2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=1 ttl=56 time=86.986 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=2 ttl=56 time=84.036 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=3 ttl=56 time=83.987 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=4 ttl=56 time=85.039 ms
2000:3:3:3:2050:79ff:fe66:6801 icmp6_seq=5 ttl=56 time=83.932 ms

PC5> ping 2000:4:4:4:2050:79ff:fe66:6802

2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=1 ttl=60 time=36.086 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=2 ttl=60 time=43.125 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=3 ttl=60 time=42.197 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=4 ttl=60 time=41.120 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=5 ttl=60 time=41.183 ms

Important Commands:

```
sh ipv6 interface brief | exclude unassigned|down
sh ipv6 protocols
sh ipv6 ospf interface
sh ipv6 ospf neighbor
sh ipv6 route ospf
sh ipv6 ospf virtual-links
```

Ratan